The Metacognitive Profile Atlas: Domain-level confidence discrimination in 33 frontier LLMs on MMLU

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https://github.com/synthiumjp/metacognitive-profile-atlas

Abstract

We release the Metacognitive Profile Atlas, a corpus of answer-plus-verbalized-confidence data from 33 frontier LLMs on 1,500 MMLU items stratified into six subject-group bins (46,157 scored observations after excluding parser-supplied values). The data allow Type-2 AUROC, the degree to which confidence discriminates correct from incorrect answers, to be decomposed by domain. The reliable finding is a population-level ordering on MMLU. Confidence discriminates best on Applied/Professional items (mean AUROC .750; ranked first in 100% of bootstrap resamples) and worst on Formal Reasoning and Natural Science (bottom two in 93% and 92%; Friedman $\chi^2(5) = 28.26$, $p < .0001$). After Holm correction Applied exceeds Formal, Science, Social and Humanities; no other pair survives, including Applied versus Factual. The ordering holds when imprecise cells are excluded or down-weighted. Per-model six-domain profiles are not reliable (median split-half $r = .16$) and are released for exploration only; the six bins are a pragmatic grouping, not a latent construct. Within-family similarity of profile shape is significant overall, but after multiplicity correction only Anthropic is individually supported. Paired-item analysis separates genuine monitoring gains (Opus 4.1 to 4.5 and 4.7) from error removal (Gemma 3 27B to Gemma 4 31B). Models whose binary-probe confidence failed or could not be certified on a separate battery discriminate well under verbalized confidence, and three Gemma 3 models show the reverse pattern. This version incorporates revisions committed during peer review, corrects descriptions of external validity-screen results, and excludes 980 parser-imputed confidence values for GPT-oss-120B, whose v1 profile was an artefact of them.

Changes in version 2. (1) GPT-oss-120B: 980 of its 1,500 confidence values were a parser default (50), not model output. Version 1 analysed them and described the result (AUROC .530, confidence SD 21.3) as an anomalous monitoring profile. They are now excluded, along with five such values in four other models and 13 unparsed answers, and all analyses, tables and figures are recomputed, with bootstrap intervals now seeded per cell; on its 518 genuine responses GPT-oss-120B has AUROC .800. (2) Claims about models being "Invalid" on a binary-probe battery relied on arXiv:2604.17707/17714v1, which contained a scoring error; corrected classifications (arXiv:2604.17714v2) are used here, and DeepSeek-R1 is not described as inverted. (3) Revisions committed during review are incorporated: claims narrowed to the population-level extremum ordering; precision-weighted, rank-stability and Holm-corrected analyses; paired-item decomposition of generational gains; per-family multiplicity correction; a parsing appendix; and the excluded MMLU subject corrected (`college_medicine`, not `elementary_mathematics`). Appendix A lists the changed claims.

Preprint.

1 Introduction

The Metacognitive Profile Atlas is a public dataset of (answer, verbalized confidence) pairs from 33 frontier large language models (LLMs) on 1,500 stratified items from the Massive Multitask Language Understanding benchmark (MMLU; Hendrycks et al., 2021). Items are grouped into six subject-group bins (Applied/Professional, Factual Recall, Formal Reasoning, Humanities/Comprehension, Natural Science, Social/Moral), 250 per bin. Models span eight families, including four Anthropic generations. After excluding parser-supplied values, the corpus holds 46,157 scored observations. Data are released under CC-BY-4.0 and code under MIT, with Croissant metadata.

Why decompose by domain. LLM confidence is used for abstention, routing and escalation [Wen et al., 2025]. Evaluations usually report one aggregate statistic, such as Type-2 AUROC (how well confidence separates correct from incorrect answers) or expected calibration error. An aggregate can hide domains where confidence carries little information. Gemini 2.5 Pro illustrates this. Its aggregate AUROC is .712, yet its confidence discriminates at .889 on Applied items and at chance (.485) on Natural Science items (§3.4).

Background. Type-2 AUROC is the standard nonparametric measure of metacognitive sensitivity in human research [Fleming and Lau, 2014, Maniscalco and Lau, 2012]: the probability that a randomly chosen correct answer receives higher confidence than a randomly chosen error. It measures discrimination, not calibration. In LLMs, verbalized and token-based confidence carry some information about correctness [Kadavath et al., 2022], but verbalized confidence is often overconfident and concentrated near the top of the scale [Xiong et al., 2023, Yang et al., 2024]; Steyvers and Peters [2025] review the area. Selective prediction, which answers only above a confidence threshold, depends on discrimination [El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017]. Mechanistic work suggests verbal confidence reflects a distinct internal evaluation [Kumaran et al., 2026], separable from calibration directions in the residual stream [Miao and Ungar, 2026]. Hidden-state signatures of known versus unknown items are also reported [Kim, 2026]. Haznitrama et al. [2026] profile capability across many models. The atlas asks a narrower question: whether confidence discrimination differs systematically across content domains, across many models, on one benchmark.

Contributions.

1. A 33-model dataset of item-level answers and verbalized confidence on 1,500 stratified MMLU items, with per-cell error counts, bootstrap intervals, parsing code and an item locator table.
2. A population-level ordering on MMLU: Applied/Professional is the domain where confidence discriminates best, and Formal Reasoning and Natural Science are where it discriminates worst. The ordering survives precision weighting, bootstrap rank analysis and Holm-corrected pairwise tests. The middle domains are not distinguishable.
3. Evidence about what is *not* reliable: per-model six-domain profiles have weak split-half reliability, and the six bins do not form cohesive subject groups. Per-model profiles and family patterns are released as exploratory.
4. A paired-item decomposition showing that equal aggregate gains between model generations can come from opposite mechanisms: genuine monitoring improvement or error removal.

The atlas is a benchmark-stage description on MMLU, not a deployment certification. All claims are specific to MMLU and to one verbalized-confidence prompt.

2 Methods

Models and items. Thirty-three LLMs from eight families were evaluated through the Kaggle Benchmarks platform (March–April 2026). The families were Anthropic (Opus 4.1–4.7, Sonnet 4–4.6, Haiku 4.5; $n = 8$), DeepSeek (V3.1, V3.2, R1; $n = 3$), Google-Gemini (2.0 Flash through 3.1 Pro; $n = 7$), Google-Gemma (3 1B through 4 31B; $n = 5$), OpenAI (GPT-5.4 nano, mini and full, GPT-oss-20B and 120B; $n = 5$), Qwen ($n = 4$) and Zhipu (GLM-5). All calls used greedy decoding (temperature 0) and an independent conversation per item. Items are 1,500 MMLU test

questions drawn deterministically (seed 42), 250 per bin. Fifty-six of 57 MMLU subjects were mapped a priori to the six bins. `college_medicine` was left unmapped because it is ambiguous between Natural Science and Applied/Professional. (Version 1 named the excluded subject as `elementary_mathematics`, which is in fact mapped to Formal Reasoning.) Table 1 summarises the bins. A per-observation (subject, split, row-index) locator table is released.

Table 1: MMLU-to-domain mapping (partial). Full mapping in the repository.

Domain bin	Label	Available items	Sampled
Applied/Professional	Applied	3,610	250
Factual Recall	Factual	2,998	250
Formal Reasoning	Formal	1,910	250
Humanities/Comprehension	Human.	987	250
Natural Science	Science	1,532	250
Social/Moral	Social	2,832	250

Elicitation and parsing. Each item was presented as multiple choice (A–D). The model was asked for its answer and a confidence from 0 to 100 in a fixed format, with no chain-of-thought cue (Appendix B). Answers and confidences were extracted by a documented rule hierarchy (Appendix C). Two kinds of record are excluded before analysis. The first is unparsed answers (13 records across six models). The second is confidence values the parser supplied because none could be extracted: the parser substituted 50, and 980 of the 985 such records belong to GPT-oss-120B, which never states 50 itself. The other five are in four models. In every affected model no response genuinely states a confidence of 50, so all its 50s can be identified as parser defaults. These values describe the pipeline, not the model, and including them in version 1 manufactured an apparently anomalous profile for GPT-oss-120B. After exclusion, 46,157 observations remain. Sixteen models have fewer than 1,500 scored items: twelve because of platform timeouts during the run (the smallest is GLM-5 with 598) and four because of the exclusions. GPT-oss-120B retains 518 items.

Analysis. Type-2 AUROC was computed per model and per model-domain cell from the raw 0–100 confidence. Bootstrap 95% intervals used 1,000 resamples per cell, each cell seeded separately so that its interval depends only on its own data, and per-cell error counts are reported (Table A6). The domain ordering was tested with Friedman's test and Kendall's W over the 33 models. Robustness was checked in four ways: excluding wide-interval or few-error cells, inverse-variance weighting, bootstrap rank stability (500 item resamples within every cell), and Holm-corrected Wilcoxon contrasts for all 15 domain pairs. Ipsative profiles subtract each model's own mean across domains. Family similarity of profile shape compares within-family and between-family correlations of ipsative profiles by permutation (10,000 shuffles), with per-family tests Holm-corrected.

3 Results

Dataset summary. Accuracy ranges from .388 (Gemma 3 1B) to .951 (Opus 4.6, Gemini 3 Flash). Aggregate AUROC ranges from .498 (Gemma 3 1B) to .806 (Opus 4.6). The 198 model-domain cells have a median of 20 errors (minimum 2; 28 cells have fewer than 10). Their bootstrap intervals have median width .202; 36% are wider than .25, mostly in high-accuracy models where few errors leave the estimate unstable. Table A2 gives the full matrix and Figure 1 the heatmap.

3.1 Domain ordering

Confidence discriminates best on Applied items and worst on Formal and Science items (Table 2, Figure 2; Friedman $\chi^2(5) = 28.26$, $p < .0001$; Kendall's $W = .171$). Applied ranks in the top two within 21 of 33 models, and Formal or Science occupies a bottom-two rank in 27 of 33. The low concordance means models agree on the extremes, not on the middle.

Robustness. The ordering does not depend on imprecise cells (Table 3). Applied stays first and Formal and Science stay in the bottom two under every exclusion rule and under inverse-variance weighting. Under the exclusions, concordance rises (W = .20 to .32), although the Friedman test then rests on fewer complete models. In 500 bootstrap resamples of items within every cell, Applied ranks first in 100% of resamples, Formal falls in the bottom two in 93% and Science in 92%. The

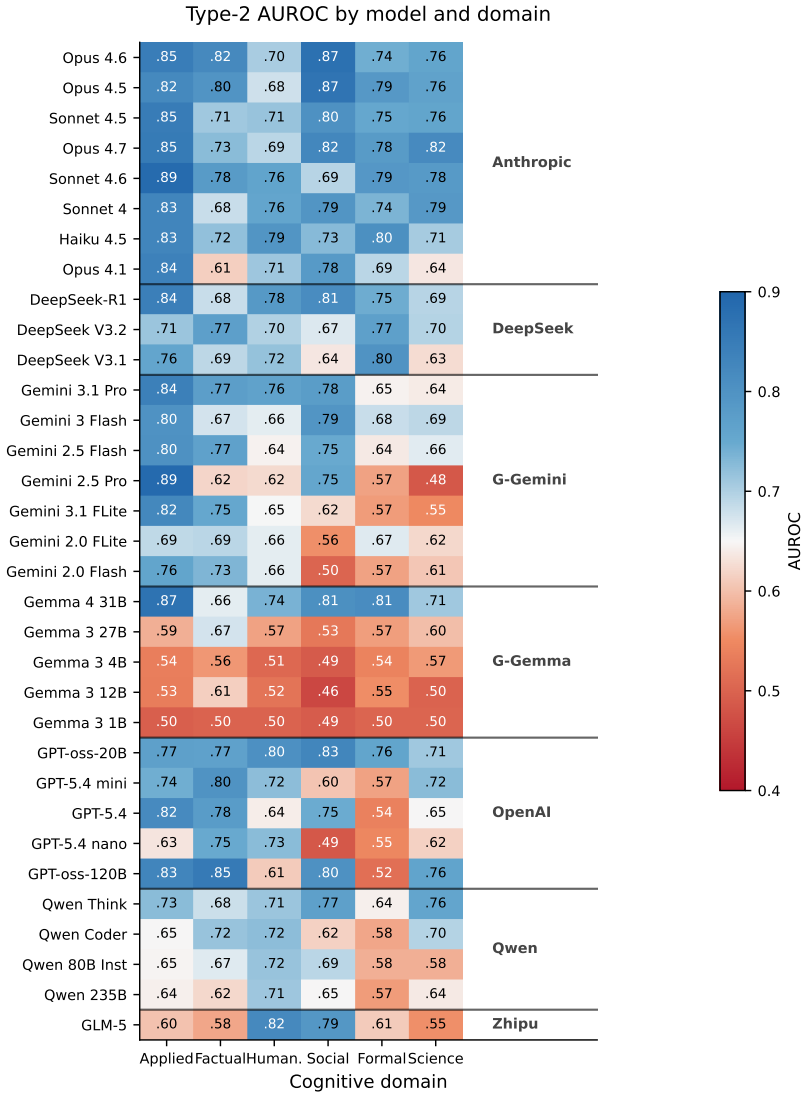


Figure 1: Type-2 AUROC by model (rows) and MMLU domain bin (columns). The colour scale spans .40 to .90, so its visual midpoint is about .65, not chance (.50). Family separators are horizontal lines. Full values: Table A2; intervals and error counts: Table A6.

Table 2: Domain means across 33 models (Type-2 AUROC).

Rank	Domain	Mean AUROC	SD	Mean rank	#1 in	Last in
1	Applied	.750	.112	2.09	15/33	0/33
2	Factual	.703	.079	3.33	7/33	5/33
3	Social	.697	.124	3.48	4/33	9/33
4	Humanities	.688	.080	3.73	4/33	4/33
5	Science	.662	.089	4.21	1/33	8/33
6	Formal	.655	.100	4.15	2/33	7/33

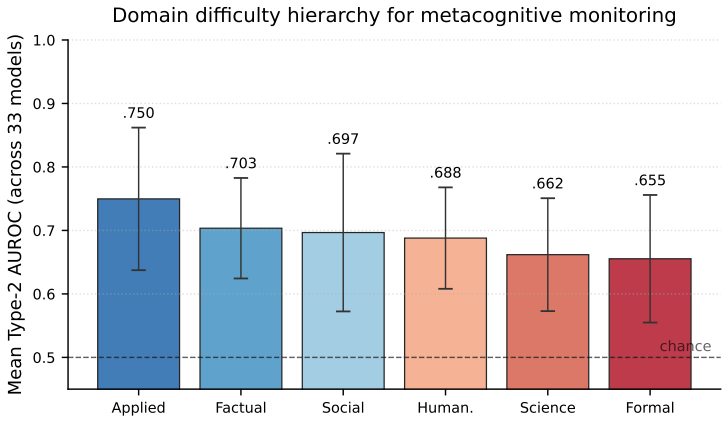


Figure 2: Mean Type-2 AUROC per domain across 33 models, with standard deviation bars.

gap between Applied and the lowest domain is .098 (95% interval [.073, .129]). Holm-corrected Wilcoxon tests over the 15 domain pairs show Applied exceeding Formal, Science, Social and Humanities ($p_{\text{Holm}} \leq .011$). No other pair survives correction, including Applied versus Factual ($p_{\text{Holm}} = .14$) and Formal versus Science ($p = .87$). Applied is the one reliably distinct domain; Formal and Science form an undifferentiated bottom pair.

Table 3: Domain ordering under precision-based variants. Friedman tests use only models with all six cells retained.

Variant	Cells	Applied	Formal	Science	Models	Friedman	W
All cells	198	.750 (1st)	.655 (6th)	.662 (5th)	33	$\chi^2 = 28.26, p < .0001$	.171
CI width $\leq .30$	157	.754 (1st)	.615 (6th)	.659 (5th)	15	$\chi^2 = 16.94, p = .005$	.226
CI width $\leq .25$	127	.754 (1st)	.586 (6th)	.623 (5th)	10	$\chi^2 = 15.83, p = .007$	.317
≥ 10 errors	170	.754 (1st)	.642 (6th)	.672 (5th)	18	$\chi^2 = 18.03, p = .003$	.200
Inverse-variance weighted	198	.711 (1st)	.572 (5th)	.534 (6th)	–	–	–

Accuracy. The ordering is not explained by domain difficulty in any simple way. Applied has the *lowest* mean accuracy (.828) and Humanities the highest (.922). Across the six domains, mean accuracy and mean AUROC correlate negatively ($\rho = -.31$), and within models the centred correlation is $r = .01$. These checks rule out a monotonic accuracy confound, not every effect of item composition. In high-accuracy cells AUROC rests on few errors, and that sparse-error regime is what the precision analyses address.

Reliability: the extremum is reliable, the profile is not. Aggregate AUROC is reproducible across random item halves (cross-model $r = .868$). The population-level contrast between Applied and the mean of Formal and Science has a split-half correlation across models of .249 (median over 50 splits; IQR .162–.343; Spearman–Brown .40), and its sign is positive in both halves in 50 of 50 splits. Full six-domain profile shapes within a model are much less reliable. Across the same splits the median correlation between halves is .157 (Spearman–Brown .27). Over 100 splits the grand median of per-model medians is .076, and 58% of models are positive. That last figure is sensitive to individual models because per-model values are spread thinly around it. The claims of this paper therefore rest on the population-level ordering; per-model profiles are released for exploration and should be read against their intervals.

Domain bins are not cohesive. Correlations of per-subject AUROC across models are no higher between subjects in the same bin than between subjects in different bins (mean $r = .173$ in both cases). The bins group MMLU subjects by pragmatic content, not by an empirically cohesive construct. The same limitation shows at the subject level. Applied subjects rank higher on average than Formal and Science subjects (mean rank 19.6 versus 28.8 and 21.1 of 47), but the difference is not significant (Mann–Whitney $p = .18$).

3.2 Families and generations (exploratory)

Family-level aggregate AUROC ranges from .573 (Gemma) to .777 (Anthropic) (Table 4). Anthropic is the most consistent family (.708–.806 across eight models). In OpenAI, the two open-weight GPT-oss models (.793 and .800) exceed GPT-5.4 (.649), mini (.651) and nano (.582). Version 1 attributed OpenAI's spread to GPT-oss-120B's low AUROC and inferred that training recipe dominates scale; that inference rested on the imputed values and is withdrawn.

Table 4: Family-level aggregate Type-2 AUROC.

Family	# models	Mean AUROC	Range
Anthropic	8	.777	.708–.806
DeepSeek	3	.740	.716–.769
Zhipu	1	.697	–
OpenAI	5	.695	.582–.800
Google-Gemini	7	.693	.623–.765
Qwen	4	.669	.621–.745
Google-Gemma	5	.573	.498–.771

Profile-shape similarity within families. Ipsative profiles correlate more within families than between them: the mean within-family r is .406 against .091 between, a difference of .316 (permutation $p = .0001$). Per family, only Anthropic is individually supported after Holm correction (Table 5). Gemini and Qwen are nominally significant and are reported as exploratory patterns. Given the weak reliability of individual profiles, these family patterns should be treated with the same caution. Figure 3 shows the profiles.

Table 5: Per-family permutation tests of profile-shape similarity (10,000 permutations; statistic = within-family minus between-family mean correlation), Holm-corrected across six families. Zhipu ($n = 1$) is not testable.

Family	n	Within r	Between r	Statistic	p	p_{Holm}
Anthropic	8	.446	.085	.360	.003	.019
Qwen	4	.482	.025	.458	.018	.092
Google-Gemini	7	.511	.232	.279	.019	.092
Google-Gemma	5	.210	−.046	.256	.066	.197
OpenAI	5	.310	.156	.154	.155	.310
DeepSeek	3	.125	.065	.060	.370	.370

Generational change: monitoring gain versus error removal. Aggregate AUROC rises across several model generations (Figure 4). The Opus line goes from .708 (4.1) to .798 (4.5) and then stays flat (.806, .792) despite continued accuracy gains. Gemma 3 27B (.565) to Gemma 4 31B (.771) is the largest single step. A higher AUROC can arise in two ways: the newer model doubts errors it still makes (better monitoring), or it stops making the confidently wrong answers that depressed the older model's AUROC (error removal). Table 6 separates the two on the items the earlier model answered wrongly with confidence of at least 70.

Table 6: Paired-item decomposition on items the earlier model answered wrongly with confidence ≥ 70. "Fixed": share answered correctly by the later model. Δconf: mean change in confidence on items still wrong (Wilcoxon signed-rank).

Pair	Items	Fixed	Δconf (still wrong)	p	Reading
Gemma 3 27B → Gemma 4 31B	321	82.6%	−2.7	.052	Error removal
Opus 4.1 → Opus 4.5	80	35.0%	−7.9	3.1×10^{-7}	Monitoring gain
Opus 4.1 → Opus 4.7	80	45.0%	−6.6	3.3×10^{-5}	Monitoring gain
Opus 4.5 → Opus 4.7	66	37.9%	+0.2	.59	Neither
DeepSeek V3.1 → V3.2	160	37.5%	−2.0	.002	Weak monitoring gain
DeepSeek V3.2 → R1	135	47.4%	−1.8	.28	Error removal

The Gemma step is mainly error removal. Gemma 4 answers 82.6% of those items correctly, its confidence on them rises (94.8 to 97.2), and on its 32 newly wrong items it reports a mean confidence

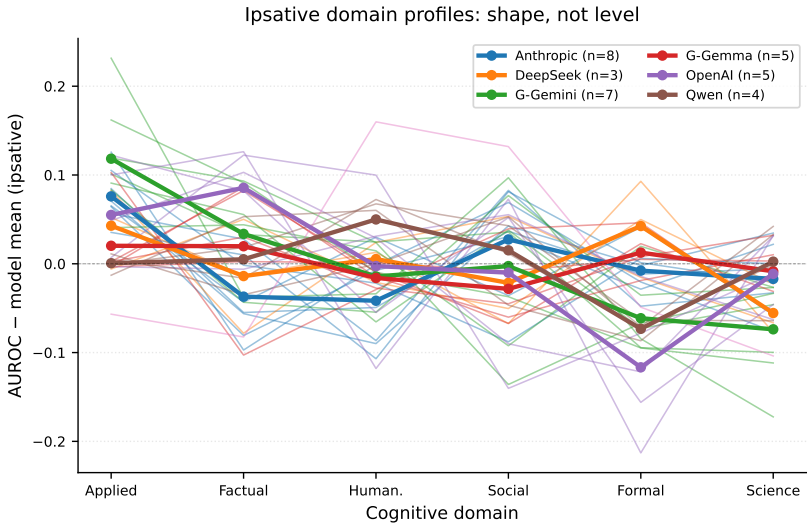


Figure 3: Ipsative domain profiles for all 33 models, coloured by family (thin lines: models; bold: family means). Centring on each model's mean isolates profile shape from level. Individual profiles are not reliable (§3.1) and are shown for exploration.

of 90.8. The Opus steps show the signature of better monitoring. Opus 4.5 lowers its confidence on the errors that persist (89.1 to 81.2) and even on the errors it fixes (85.7 to 77.9), and Opus 4.7's 31 newly wrong items carry a mean confidence of only 71.5. Similar aggregate gains can therefore come from opposite mechanisms, which is the main reason for releasing paired item-level data. Gemma 3 1B's domain intervals are the narrowest in the atlas (all widths at most .06), so its near-chance aggregate (.498) is a precise estimate of absent discrimination, not noise.

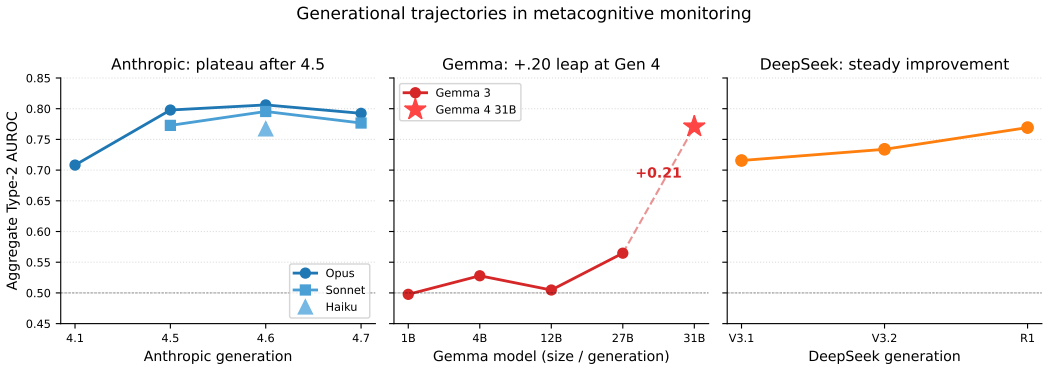


Figure 4: Aggregate Type-2 AUROC across generations for Anthropic, Gemma and DeepSeek. Table 6 decomposes the changes.

3.3 Binary-probe validity screening and verbalized confidence

Twenty of the 33 models were also tested on a separate 524-item battery that asks after each answer whether to KEEP or WITHDRAW it [Cacioli, 2026a]. A portable validity screen [Cacioli, 2026b] classifies a binary confidence signal from the correctness-by-confidence table. It is Invalid when a response set is established: confidence kept on at least 95% of errors (L), or withdrawn on at least half of correct answers (Fp), with interval guards. It is Indeterminate when such a pattern cannot be certified, often because there are too few errors, or when confidence does not measurably track correctness. It is Valid otherwise. The corrected classification (arXiv:2604.17714v2) uses the battery's five keyed tracks. It finds DeepSeek-R1 Invalid: it withdraws 95% of its correct answers, and with only six errors whether its withdrawals track correctness cannot be assessed. Gemini 3.1

Pro and Qwen3 Next 80B Thinking are Indeterminate: they keep confidence on 96% of their errors but make 24 and 27 errors, too few to certify blanket confidence. Gemini 3 Flash and GPT-5.4 nano are also Indeterminate, with no detectable item sensitivity. Version 1 of this paper described R1 as showing catastrophic inversion; that description came from a scoring error in the battery analysis and is withdrawn.

Under verbalized confidence the picture differs in both directions (Figure 5). R1, Gemini 3.1 Pro and Qwen Thinking discriminate well on MMLU (aggregate AUROC .769, .765 and .745; domain ranges .682–.843, .640–.841 and .644–.768). Gemma 3 1B, 12B and 27B are Valid on the binary battery, yet their verbalized confidence barely discriminates on MMLU (.498, .505 and .565). Across the 20 models, battery AUROC does not predict atlas AUROC (Spearman $\rho = .00$). Screened on MMLU at a cut point of 90 (confidence ≥ 90 counts as high), the four Gemma 3 models are Invalid, Gemini 2.5 Pro is Indeterminate, and the other 28 are Valid. A model's confidence quality is therefore specific to the probe format and task under which it is measured. A binary-probe result says little about verbalized confidence on another benchmark, and the reverse.

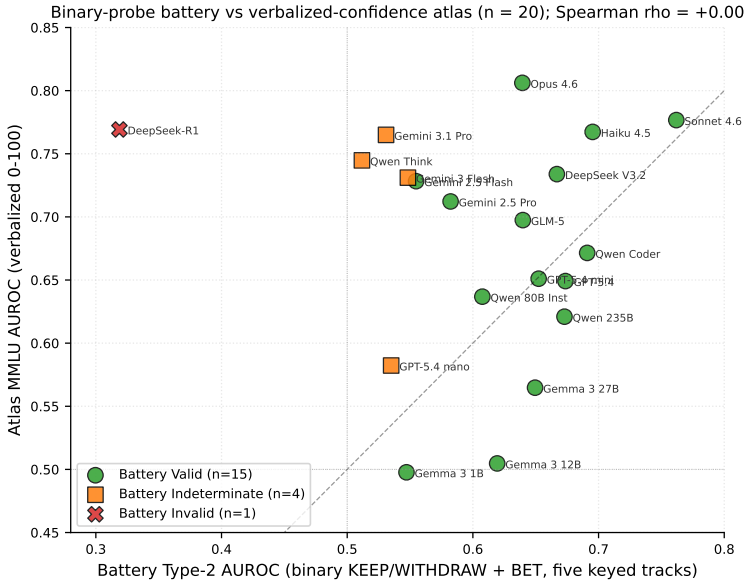


Figure 5: Battery Type-2 AUROC (binary KEEP/WITHDRAW + BET probes, five keyed tracks; Cacioli, 2026b) against atlas aggregate AUROC (verbalized 0–100 confidence on MMLU) for the 20 models in both. Markers show the corrected battery tier. The dashed line is identity.

3.4 Individual profiles

These descriptions are exploratory (§3.1).

Gemini 2.5 Pro has the widest within-model spread: Applied .889 [.794, .960], Science .485 [.470, .497], Formal .573 [.478, .778]. The Applied lower bound exceeds the Science upper bound, so the difference is not sampling noise. The aggregate (.712) gives no sign that its Science-domain confidence is at chance. This run is partial (981 items).

GLM-5 is the only model whose Humanities AUROC (.818 [.706, .972]) is well above its Applied AUROC (.602). Its run is partial (598 items), and its Humanities estimate rests on three errors. A complete rerun three months later shows what can and cannot be concluded. On the 598 items common to both runs, correctness agreed on 98.5% of items, item-level confidence correlated $r = .71$, and the six-domain ordering correlated $\rho = .77$ ($p = .072$, six domains). Humanities again exceeded Applied (.792 versus .639), and the aggregate AUROC rose from .697 to .759. The rerun's errors per domain numbered between 2 and 12, so its domain estimates are imprecise as well (Appendix I).

GPT-oss-120B. Version 1 reported this model as having the highest confidence variability in the atlas with near-chance discrimination (SD 21.3, AUROC .530), and offered it as a distinct failure

mode. That profile was produced by 980 parser-imputed values of 50. On its 518 genuine responses its aggregate AUROC is .800 and its domain values range from .515 (Formal) to .855 (Factual). Its imputation rate also differs by domain (59% to 74%; $\chi^2(5) = 18.67$, $p = .002$), so the retained items are not a random sample. With 65 to 102 items per domain and wide intervals (Formal: [.128, .918]), its profile should not be interpreted.

3.5 Validation summary

Aggregate AUROC rankings are reproducible on half the items ($r = .868$; Appendix Figure A1). The population-level extremum contrast is reproducible in sign but modest in reliability. Per-model profiles are not reliable, and the domain bins are not cohesive (§3.1). The paper's claims are limited accordingly.

4 Discussion

What the atlas shows. On MMLU, verbalized confidence across 33 frontier models discriminates correct from incorrect answers best on applied professional knowledge and worst on formal reasoning and natural science. The pattern holds at the population level under precision weighting, bootstrap rank analysis and multiplicity-corrected contrasts. One plausible account: Applied items (law, medicine, accounting) tend to have answers anchored in specific retrieved knowledge, so whether a model knows the answer may leave a clearer internal trace. Formal items admit plausible intermediate steps that end at wrong answers. This fits the hypothesis of Kim [2026] that well-learned factual material produces more separable internal states. Nothing here examines hidden states, so the account is a hypothesis, not a mechanism.

What it does not show. It does not show that any individual model has a stable domain profile: per-model profiles have weak split-half reliability. It does not show that the six bins are cognitive constructs: subjects within a bin are no more alike than subjects across bins. It does not show that the ordering holds beyond MMLU or beyond this prompt; replication on a benchmark with non-overlapping items (for example GPQA) is the necessary next test. Type-2 AUROC measures discrimination, not calibration.

Implications for confidence-based abstention. If a deployment applies one confidence threshold across content types, the population-level result predicts that the threshold will be less informative on formal and scientific content than on applied professional content. This argues for checking discrimination separately on the content a system will see, rather than relying on a single aggregate. It does not support looking up a particular model's domain profile in this atlas and treating it as reliable. In flagship models, adjacent-domain differences are not interpretable at current item counts: GPT-5.4's Formal AUROC is .539 [.452, .631], for example. Those comparisons need many more errors per cell.

Probe format. Classifications from a binary KEEP/WITHDRAW probe did not predict discrimination under verbalized confidence, in either direction (§3.3). Statements about a model's confidence should name the probe and task: "no detectable signal under probe X on task Y", not "no signal".

Parsing is part of the measurement. The largest correction in this version came from parser-supplied values being treated as model output. Releasing raw outputs, extraction code and per-model fallback counts made the error findable. We recommend that verbalized-confidence datasets report imputation and fallback rates per model, and exclude imputed values before analysis.

Limitations. The results cover MMLU only and one verbalized-confidence prompt, with greedy decoding. The domain mapping is a priori and not empirically cohesive. Per-cell estimates are imprecise for high-accuracy models, with 36% of intervals wider than .25 and 28 cells with fewer than 10 errors. Sixteen runs are partial (twelve from platform timeouts, four from exclusions), and GPT-oss-120B's retained items are a non-random subset. Answers and confidences were extracted by rules, with fallbacks that can occasionally capture an unintended letter or number (Appendix C). No causal mechanism is tested.

5 Conclusion

Aggregate confidence metrics hide a reliable population-level pattern on MMLU: verbalized confidence in frontier models discriminates best on applied professional knowledge and worst on formal reasoning and natural science. Per-model profiles are too unreliable to interpret individually at this item count. The atlas releases the item-level data, error counts, parsing code and locators needed to test these patterns on other benchmarks and probes.

Open science

Data (item-level CSVs, error counts, bootstrap intervals, MMLU locators, parsing fallback counts), extraction and analysis code, and a single script that regenerates every number and figure are at `https://github.com/synthiumjp/metacognitive-profile-atlas` (MIT, code) and `https://huggingface.co/datasets/synthiumjp/metacognitive-profile-atlas` (CC-BY-4.0, data; Croissant metadata).

Use of AI

Claude (Anthropic) assisted with benchmark design, analysis code and manuscript preparation. Analysis code was checked against `scikit-learn` reference implementations and by re-running the full pipeline from raw data; the corrections in this version were found by such re-analysis. All scientific decisions, the domain mapping and the interpretations are the author's.

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A Changes from version 1

Table A1 lists every claim in version 1 that this version withdraws or materially changes. Version 1 remains available on arXiv for the record.

B Elicitation prompt template

```
Answer this multiple-choice question.

{question}

First, state your answer (A, B, C, or D).
Then, state your confidence that your answer is correct as a number from 0
(pure guess) to 100 (certain).

Reply in EXACTLY this format:
Answer: [letter]
Confidence: [number]
```

Each (model, item) pair used an independent conversation context. Greedy decoding (temperature 0) throughout.

C Answer and confidence parsing

Answers and confidences were extracted from each response by fixed rules, applied in order with the first match winning. The code is released (`notebooks/atlas_benchmark.py`), and a verification script re-runs it over the raw platform outputs and reproduces all 47,151 records.

Answer. (1) `Answer:\s*([A-Da-d])`; (2) otherwise the first standalone letter A–D; (3) otherwise the answer is recorded as X. Thirteen responses (0.03%) were X: GPT-oss-120B 5, Gemini 3.1 Flash-Lite 2, Gemma 3 27B 2, Haiku 4.5 2, Gemma 3 12B 1, Qwen3 Next 80B Instruct 1. Version 1 scored them as incorrect; version 2 excludes them. The fallback rule fired on 42 responses, 28 of them from Gemini 2.5 Flash. A first standalone letter can occasionally be an unintended one (for example the article "A"); spot checks found the intended answer.

Confidence. (1) `Confidence:\s*(\d+)`, clipped to 0–100; (2) otherwise the first standalone integer of up to three digits in 0–100; (3) otherwise the value 50 is imputed. Rule 3 fired on 985 responses: 980 from GPT-oss-120B (65% of its responses) and five from four other models. The

Table A1: Claims changed between version 1 and version 2.

Version 1	**Version 2**
GPT-oss-120B shows the highest confidence variability (SD 21.3) with near-chance discrimination (AUROC .530) and below-chance Science monitoring, a distinct failure mode.	Withdrawn. 980 of its 1,500 confidence values were the parser default of 50. On its 518 genuine responses AUROC is .800. The retained items are a non-random subset, and the domain profile is not interpreted (§3.4, Appendix E).
Within OpenAI, GPT-oss-20B outperforms the six-times-larger GPT-oss-120B, so training recipe dominates scale.	Withdrawn; it rested on the imputed values.
DeepSeek-R1, Gemini 3.1 Pro and Qwen Think are battery-Invalid; R1 shows catastrophic inversion on the binary probe.	Corrected from arXiv:2604.17714v2. R1 is Invalid because it withdraws 95% of its correct answers; it is not inverted. Gemini 3.1 Pro and Qwen Think are Indeterminate (blanket confidence suspected but not certifiable on 24 and 27 errors).
Tier-level assignments are preserved across probe formats in every battery-Valid case.	Withdrawn. Battery AUROC does not predict atlas AUROC ($\rho = .00$), and three battery-Valid Gemma 3 models barely discriminate on MMLU (§3.3).
Introduction: every Valid model in a 20-model battery sample had a domain AUROC below .55 [Cacioli, 2026a].	Removed; it relied on the battery scoring corrected in arXiv:2604.17714v2.
The Gemma 3 $\rightarrow$ Gemma 4 step shows that a single generational step can recover an entire class of failure modes.	Reframed. The paired-item decomposition shows that step is mainly error removal, not improved monitoring (Table 6).
Profile-level split-half: grand median $r = .184$, 61% of models positive; aggregate split-half $r = .893$.	Recomputed: grand median $r = .076$, 58% positive; aggregate $r = .868$. Claims rest on the population-level extremum ordering.
Friedman $\chi^2(5) = 27.04$, $W = .164$; 47,151 observations.	$\chi^2(5) = 28.26$, $W = .171$; 46,157 observations. The ordering is unchanged.
The excluded MMLU subject is `elementary_mathematics`.	The excluded subject is `college_medicine`; `elementary_mathematics` is mapped to Formal Reasoning. The analysis was always run with the correct mapping; only the text was wrong.
Cross-study comparison with MMLU AUROCs computed from median-binarised confidence.	Removed. Median binarisation is superseded in the validity screen [Cacioli, 2026b], whose verdicts depend on tie handling.
Figure 1 colour scale described as diverging around chance (.50).	The scale spans .40–.90, so its visual midpoint is about .65. The caption now says so.
"Confidence reliability", "monitoring quality" used for Type-2 AUROC.	"Confidence discrimination". Type-2 AUROC measures discrimination, not calibration or reliability in the psychometric sense.
Bootstrap intervals drawn from one random stream across all cells (seed 42).	Each cell seeded separately, so an interval depends only on its own data. Interval endpoints shift by up to a few hundredths.

raw outputs show that none of these five models ever states a confidence of 50, so each 50 in their series is a parser default. All 985 are excluded in version 2 (some coincide with X answers). Per-model counts for every route are released (`data/parsing_fallback_counts.csv`).

Reasoning traces. For DeepSeek-R1, whose output includes its reasoning, rule 1 captures the first "Confidence:" in the response, which can fall inside the reasoning rather than in the final answer. On 82 items (5.5%) the captured value differs from the final stated confidence (mean 82.0 versus 87.4). Using the final values changes R1's aggregate AUROC from .769 to .766. The released data keep the values as originally extracted, and the discrepant items are listed in `data/thinking_conf_deepseek-r1-0528.csv`.

Models and runs. Gemma 4 26B A4B appears in the benchmark's model list but its run never started (no logs or outputs), so no data exist for it. No model was excluded on the basis of its responses.

D Full 33-model AUROC matrix

Table A2 reports Type-2 AUROC for every model-domain cell. Figure 1 shows the same values.

Table A2: Type-2 AUROC per model-domain cell, all 33 models. n = scored items after excluding unparsed answers and parser-imputed confidences (16 models have fewer than 1,500; GPT-oss-120B† retains 518 after 982 exclusions and should not be interpreted at domain level). Agg = AUROC over all of a model's items. Models grouped by family, sorted by Agg within family.

Model	n	Applied	Factual	Human.	Social	Formal	Science	Agg
Opus 4.6	1500	.847	.818	.704	.873	.743	.758	.806
Opus 4.5	1500	.821	.795	.678	.867	.787	.765	.798
Sonnet 4.5	1500	.849	.709	.715	.801	.752	.758	.795
Opus 4.7	1499	.852	.726	.692	.825	.780	.816	.792
Sonnet 4.6	1500	.887	.780	.760	.694	.788	.782	.777
Sonnet 4	1500	.828	.683	.756	.794	.735	.786	.773
Haiku 4.5	1498	.829	.720	.793	.727	.805	.706	.767
Opus 4.1	1500	.838	.615	.711	.781	.691	.636	.708
DeepSeek-R1	1500	.843	.682	.785	.814	.745	.694	.769
DeepSeek V3.2	1371	.713	.769	.698	.669	.769	.696	.734
DeepSeek V3.1	1380	.756	.690	.717	.636	.797	.626	.716
Gemini 3.1 Pro	1346	.841	.773	.764	.776	.645	.640	.765
Gemini 3 Flash	1500	.800	.673	.662	.794	.681	.690	.731
Gemini 2.5 Flash	1360	.801	.765	.644	.748	.637	.664	.728
Gemini 2.5 Pro	981	.889	.619	.623	.754	.573	.485	.712
Gemini 3.1 FLite	1365	.823	.752	.651	.624	.566	.549	.655
Gemini 2.0 FLite	1500	.688	.692	.662	.555	.667	.621	.639
Gemini 2.0 Flash	1500	.758	.732	.663	.503	.572	.606	.623
Gemma 4 31B	1500	.869	.663	.737	.806	.812	.710	.771
Gemma 3 27B	1497	.585	.674	.573	.528	.569	.598	.565
Gemma 3 4B	1445	.536	.564	.506	.489	.541	.566	.528
Gemma 3 12B	1446	.533	.613	.522	.464	.554	.500	.505
Gemma 3 1B	1499	.495	.501	.500	.490	.503	.502	.498
GPT-oss-120B†	518	.829	.855	.611	.801	.515	.760	.800
GPT-oss-20B	1500	.769	.766	.803	.827	.758	.709	.793
GPT-5.4 mini	1356	.742	.797	.723	.604	.572	.725	.651
GPT-5.4	1500	.818	.778	.640	.747	.539	.649	.649
GPT-5.4 nano	1500	.633	.748	.726	.486	.548	.615	.582
Qwen Think	1500	.726	.680	.714	.768	.644	.757	.745
Qwen Coder	1500	.650	.716	.723	.616	.576	.695	.671
Qwen 80B Inst	1498	.648	.667	.716	.691	.585	.584	.637
Qwen 235B	1500	.644	.622	.711	.650	.567	.637	.621
GLM-5	598	.602	.576	.818	.790	.609	.555	.697

E GPT-oss-120B

GPT-oss-120B produced a parseable confidence on 520 of 1,500 items, and 518 remain after the two further unparsed answers are excluded. The imputation rate differs by domain, from 59% (Natural Science) to 74% (Factual Recall) ($\chi^2(5) = 18.67$, $p = .002$), so the retained items are not a random sample of the item pool. Accuracy is similar on imputed and retained items (.900 and .896), but whether the model states a confidence at all may depend on item properties that also bear on discrimination. Its aggregate AUROC (.800) and domain values (Table A2) are reported for completeness. With 65–102 items per domain and intervals as wide as [.128, .918] (Formal), its domain profile should not be interpreted. Excluding the model entirely leaves the domain ordering unchanged (Friedman $\chi^2(5) = 26.43$, $p < .0001$, $n = 32$).

F Robustness analyses

Pairwise contrasts. Table A3 gives Wilcoxon signed-rank tests over the 33 models for all 15 domain pairs, Holm-corrected.

Table A3: Pairwise domain contrasts (Wilcoxon signed-rank over 33 models; Holm correction across 15 pairs). Diff = mean AUROC difference, first minus second.

Pair	Diff	p	p_{Holm}
Applied vs Science	.088	$< .0001$	$< .0001$
Applied vs Formal	.094	$< .0001$	$< .0001$
Applied vs Social	.053	.0003	.005
Applied vs Humanities	.062	.001	.011
Factual vs Science	.042	.005	.057
Applied vs Factual	.046	.014	.144
Factual vs Formal	.048	.035	.318
Formal vs Social	−.041	.044	.355
Science vs Social	−.035	.060	.420
Formal vs Humanities	−.033	.063	.420
Humanities vs Science	.026	.135	.674
Factual vs Humanities	.016	.280	1.000
Humanities vs Social	−.009	.597	1.000
Factual vs Social	.007	.846	1.000
Formal vs Science	−.007	.874	1.000

Rank stability. Items were resampled with replacement within every model-domain cell (500 resamples), and domain means were re-ranked each time. Applied ranked first in 100% of resamples. The probability of a bottom-two rank was .932 for Formal, .924 for Science, .132 for Humanities, .008 for Social, .004 for Factual and 0 for Applied.

Error counts. AUROC in a cell with few errors is unstable. Cells have a median of 20 errors (minimum 2); 28 cells have fewer than 10 and 5 have fewer than 5. Per-cell counts are in Table A6.

Subject level. For the 47 subjects with at least 30 errors pooled across models, the seven Applied subjects have a mean AUROC rank of 19.6, against 28.8 for 11 Formal subjects and 21.1 for nine Science subjects. Applied subjects do not rank significantly higher than Formal and Science subjects combined (one-sided Mann–Whitney $U = 87$, $p = .18$). The domain-level ordering is a property of the bins in aggregate, not a uniform property of their subjects.

G Reliability

Items were randomly split into halves, and AUROC was computed on each half for every model (Figure A1). Aggregate AUROC correlates $r = .868$ across models between halves. For the population-level extremum contrast (Applied minus the mean of Formal and Science), the cross-model split-half correlation has median .249 over 50 splits (IQR .162–.343), and its population mean is positive in both halves in all 50. For full six-domain profiles, the within-model correlation between halves has median .157 over the same splits. Over 100 splits the grand median of per-model medians is .076,

and 58% of models are positive. These estimates move by several hundredths when a handful of records change, which is itself a sign of their imprecision.

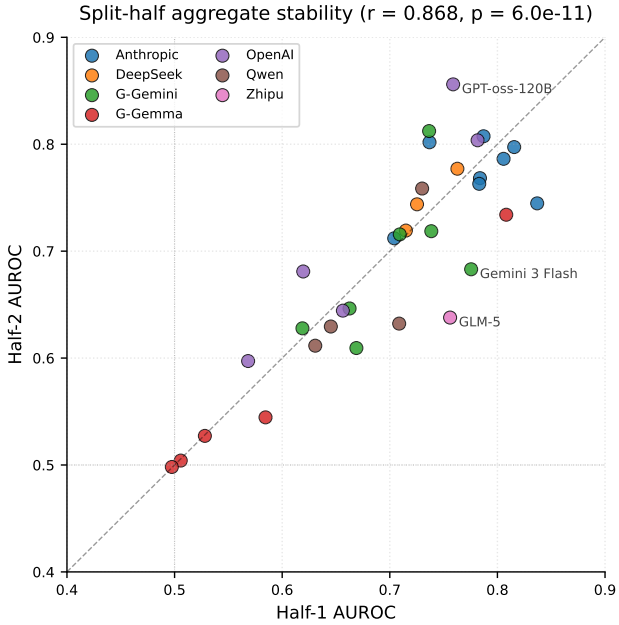


Figure A1: Split-half stability of aggregate Type-2 AUROC across 33 models. Each point is one model; axes are AUROC on two random halves of its items. Cross-model $r = .868$.

H Families and generations

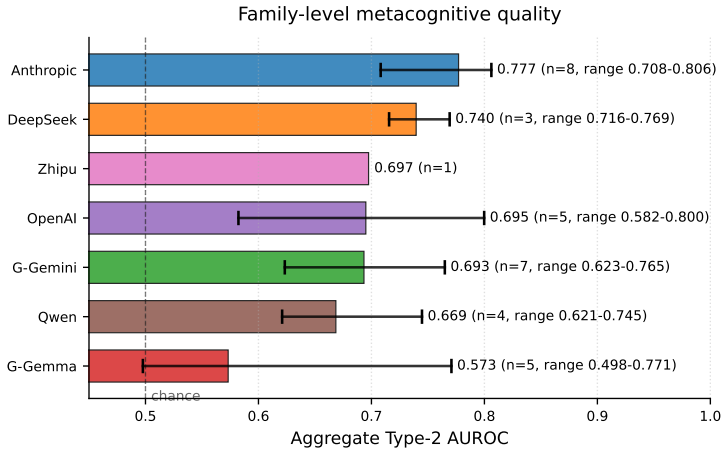


Figure A2: Family mean domain profiles (Type-2 AUROC). Exploratory; see §3.1 on profile reliability.

Item-level transitions. Table A4 cross-classifies shared items by correctness under the earlier and later model in three generational pairs. Per-item confidence correlates more weakly across the Gemma step (Spearman $\rho = .36$) than across Anthropic steps (.75–.85) or DeepSeek steps (.66–.68).

Table A4: Correctness transitions on shared items (W = wrong, C = correct; earlier → later). The last column is the later model's mean confidence on items it newly gets wrong.

Pair	W→W	W→C	C→W	C→C	Conf. on C→W
Gemma 3 27B → Gemma 4 31B	57	265	32	1,144	90.8
Opus 4.1 → Opus 4.7	49	37	31	1,382	71.5
DeepSeek V3.2 → R1	71	64	48	1,188	84.6

I GLM-5 retest

GLM-5's original run stopped after 598 items. It was rerun on the full item set three months later under the same prompt and settings. Table A5 compares the two runs on the 598 shared items. The six domain values correlate $\rho = .77$ ($p = .072$) and $r = .75$. Correctness agrees on 98.5% of items, and item-level confidence correlates $r = .71$. In both runs Humanities exceeds Applied. With 2 to 14 errors per domain, neither run's domain estimates are precise.

Table A5: GLM-5 on the same 598 items in two runs three months apart: errors and Type-2 AUROC per domain.

		Original run		Rerun	
Domain	n	**Errors**	**AUROC**	**Errors**	**AUROC**
Applied	84	4	.602	5	.639
Factual	111	8	.576	10	.702
Formal	102	3	.609	5	.757
Humanities	91	3	.818	2	.792
Science	110	3	.555	4	.709
Social	100	14	.790	12	.777
Aggregate	598	35	.697	38	.759

J Partial runs

Twelve runs are incomplete because of platform timeouts; the smallest are GLM-5 (598 items) and Gemini 2.5 Pro (981). Four further models fall below 1,500 only because of the exclusions in Appendix C. Whether an item is present was tested against its domain for each incomplete model (χ^2, 5 df). Presence is unrelated to domain ($p > .07$) for every model except Gemma 3 12B ($p = .010$; 54 items missing) and GPT-oss-120B ($p = .003$; Appendix E).

K Future directions

Transport. Whether the population-level ordering holds on a benchmark with non-overlapping items (for example GPQA) and under other elicitation formats is the main open question.

Mechanism. Hidden-state analyses of open-weight models (Gemma 4 31B, GPT-oss-20B, Qwen Think) could test whether Applied items produce more separable internal states than Formal items.

Precision. Per-model profiles would need many more errors per cell to be interpretable. For high-accuracy models that means many more items per domain, or harder items.

Repair. Whether targeted fine-tuning with confidence in the training signal can raise discrimination in a model's weakest domain is untested. The Gemma step shows that a large aggregate gain can come from error removal alone (Table 6), so any such test should decompose its effect the same way.

L Procedural analogy to clinical assessment

Clinical neuropsychological assessment follows a fixed interpretive sequence. Validity indicators are checked first [Larrabee, 2012]. If the protocol is valid, aggregate scores give an overall level,

and the profile of index scores is then examined for relative strengths and weaknesses. The atlas follows the same order: screen the confidence signal [Cacioli, 2026b], compute the aggregate, then examine the domain profile. The analogy is procedural only. Human cognitive domains rest on extensive factor-analytic validation; the MMLU bins used here do not, and §3.1 shows they are not cohesive. The analogy also marks a limit: in clinical practice an index score is interpreted only when its reliability supports it, and at current item counts most per-model domain profiles here do not meet that standard.

M Bootstrap intervals and error counts

Table A6 gives bootstrap 95% intervals and error counts for all 198 model-domain cells. Median interval width is .202.

Table A6: Bootstrap 95% confidence intervals and error counts for all 198 model-domain AUROC cells (1,000 resamples per cell, seeded per cell). Median CI width = .202. Archived as `data/atlas_bootstrap_cis.csv` and `data/atlas_cell_error_counts.csv`.

Model	Domain	n	Errors	AUROC	CI lo	CI hi
Opus 4.6	Applied	250	18	0.847	0.765	0.923
Opus 4.6	Factual	250	15	0.818	0.694	0.927
Opus 4.6	Human.	250	8	0.704	0.421	0.940
Opus 4.6	Social	250	12	0.873	0.800	0.938
Opus 4.6	Formal	250	8	0.743	0.446	0.977
Opus 4.6	Science	250	13	0.758	0.623	0.886
Opus 4.5	Applied	250	18	0.821	0.736	0.893
Opus 4.5	Factual	250	16	0.795	0.657	0.907
Opus 4.5	Human.	250	8	0.678	0.409	0.919
Opus 4.5	Social	250	14	0.867	0.759	0.946
Opus 4.5	Formal	250	10	0.787	0.565	0.950
Opus 4.5	Science	250	13	0.764	0.629	0.884
Sonnet 4.5	Applied	250	26	0.849	0.781	0.910
Sonnet 4.5	Factual	250	13	0.709	0.528	0.857
Sonnet 4.5	Human.	250	7	0.715	0.464	0.915
Sonnet 4.5	Social	250	16	0.801	0.708	0.889
Sonnet 4.5	Formal	250	7	0.752	0.507	0.989
Sonnet 4.5	Science	250	12	0.758	0.593	0.925
Opus 4.7	Applied	250	19	0.853	0.764	0.917
Opus 4.7	Factual	249	14	0.726	0.581	0.868
Opus 4.7	Human.	250	7	0.692	0.389	0.970
Opus 4.7	Social	250	17	0.825	0.735	0.895
Opus 4.7	Formal	250	12	0.780	0.592	0.943
Opus 4.7	Science	250	11	0.817	0.677	0.934
Sonnet 4.6	Applied	250	20	0.887	0.822	0.940
Sonnet 4.6	Factual	250	17	0.780	0.646	0.903
Sonnet 4.6	Human.	250	13	0.760	0.578	0.916
Sonnet 4.6	Social	250	27	0.694	0.598	0.784
Sonnet 4.6	Formal	250	8	0.788	0.540	0.976
Sonnet 4.6	Science	250	12	0.782	0.629	0.906
Sonnet 4	Applied	250	21	0.828	0.766	0.888
Sonnet 4	Factual	250	16	0.683	0.543	0.815
Sonnet 4	Human.	250	13	0.756	0.616	0.873
Sonnet 4	Social	250	21	0.794	0.698	0.866
Sonnet 4	Formal	250	9	0.735	0.558	0.915
Sonnet 4	Science	250	16	0.786	0.665	0.893
Haiku 4.5	Applied	248	28	0.829	0.743	0.907

Continued on next page

Table A6 – continued

Model	Domain	n	Errors	AUROC	CI lo	CI hi
Haiku 4.5	Factual	250	26	0.720	0.610	0.819
Haiku 4.5	Human.	250	11	0.793	0.645	0.915
Haiku 4.5	Social	250	28	0.728	0.638	0.810
Haiku 4.5	Formal	250	17	0.805	0.663	0.923
Haiku 4.5	Science	250	16	0.706	0.567	0.838
Opus 4.1	Applied	250	18	0.838	0.781	0.889
Opus 4.1	Factual	250	11	0.615	0.444	0.775
Opus 4.1	Human.	250	7	0.711	0.479	0.924
Opus 4.1	Social	250	20	0.781	0.682	0.867
Opus 4.1	Formal	250	12	0.691	0.521	0.840
Opus 4.1	Science	250	18	0.636	0.526	0.761
DeepSeek-R1	Applied	250	27	0.843	0.775	0.907
DeepSeek-R1	Factual	250	20	0.682	0.539	0.815
DeepSeek-R1	Human.	250	14	0.785	0.635	0.922
DeepSeek-R1	Social	250	33	0.814	0.725	0.889
DeepSeek-R1	Formal	250	13	0.745	0.596	0.875
DeepSeek-R1	Science	250	20	0.694	0.540	0.830
DeepSeek V3.2	Applied	228	31	0.713	0.612	0.803
DeepSeek V3.2	Factual	221	23	0.769	0.683	0.849
DeepSeek V3.2	Human.	229	13	0.698	0.562	0.817
DeepSeek V3.2	Social	231	36	0.669	0.576	0.751
DeepSeek V3.2	Formal	231	13	0.769	0.607	0.901
DeepSeek V3.2	Science	231	19	0.697	0.557	0.811
DeepSeek V3.1	Applied	234	44	0.756	0.681	0.820
DeepSeek V3.1	Factual	229	22	0.690	0.578	0.799
DeepSeek V3.1	Human.	230	20	0.717	0.591	0.825
DeepSeek V3.1	Social	225	38	0.636	0.531	0.728
DeepSeek V3.1	Formal	228	24	0.797	0.691	0.888
DeepSeek V3.1	Science	234	29	0.626	0.514	0.733
Gemini 3.1 Pro	Applied	227	13	0.841	0.715	0.949
Gemini 3.1 Pro	Factual	215	14	0.773	0.629	0.899
Gemini 3.1 Pro	Human.	228	8	0.764	0.533	0.972
Gemini 3.1 Pro	Social	225	19	0.776	0.663	0.883
Gemini 3.1 Pro	Formal	230	6	0.645	0.469	0.891
Gemini 3.1 Pro	Science	221	9	0.640	0.467	0.826
Gemini 3 Flash	Applied	250	13	0.800	0.718	0.881
Gemini 3 Flash	Factual	250	13	0.673	0.516	0.825
Gemini 3 Flash	Human.	250	9	0.662	0.496	0.829
Gemini 3 Flash	Social	250	17	0.794	0.693	0.879
Gemini 3 Flash	Formal	250	9	0.681	0.504	0.862
Gemini 3 Flash	Science	250	12	0.690	0.530	0.845
Gemini 2.5 Flash	Applied	227	28	0.801	0.709	0.880
Gemini 2.5 Flash	Factual	219	20	0.765	0.638	0.883
Gemini 2.5 Flash	Human.	231	12	0.644	0.485	0.807
Gemini 2.5 Flash	Social	228	32	0.748	0.661	0.835
Gemini 2.5 Flash	Formal	230	19	0.637	0.534	0.745
Gemini 2.5 Flash	Science	225	16	0.664	0.546	0.800
Gemini 2.5 Pro	Applied	165	17	0.889	0.794	0.960
Gemini 2.5 Pro	Factual	172	11	0.619	0.484	0.788
Gemini 2.5 Pro	Human.	160	6	0.623	0.449	0.844
Gemini 2.5 Pro	Social	150	15	0.754	0.623	0.893
Gemini 2.5 Pro	Formal	163	6	0.573	0.478	0.777
Gemini 2.5 Pro	Science	171	7	0.485	0.469	0.497
Gemini 3.1 FLite	Applied	231	28	0.823	0.737	0.898
Gemini 3.1 FLite	Factual	233	18	0.752	0.636	0.858

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Table A6 – continued

Model	Domain	n	Errors	AUROC	CI lo	CI hi
Gemini 3.1 FLite	Human.	222	9	0.651	0.471	0.841
Gemini 3.1 FLite	Social	229	29	0.624	0.531	0.714
Gemini 3.1 FLite	Formal	227	31	0.566	0.506	0.641
Gemini 3.1 FLite	Science	223	19	0.549	0.474	0.646
Gemini 2.0 FLite	Applied	250	56	0.688	0.623	0.749
Gemini 2.0 FLite	Factual	250	35	0.692	0.585	0.777
Gemini 2.0 FLite	Human.	250	25	0.662	0.547	0.769
Gemini 2.0 FLite	Social	250	48	0.555	0.468	0.646
Gemini 2.0 FLite	Formal	250	61	0.667	0.597	0.740
Gemini 2.0 FLite	Science	250	45	0.621	0.533	0.711
Gemini 2.0 Flash	Applied	250	45	0.758	0.691	0.822
Gemini 2.0 Flash	Factual	250	29	0.732	0.629	0.822
Gemini 2.0 Flash	Human.	250	13	0.663	0.512	0.808
Gemini 2.0 Flash	Social	250	41	0.503	0.420	0.583
Gemini 2.0 Flash	Formal	250	43	0.572	0.498	0.648
Gemini 2.0 Flash	Science	250	30	0.606	0.505	0.696
Gemma 4 31B	Applied	250	17	0.869	0.773	0.942
Gemma 4 31B	Factual	250	19	0.663	0.535	0.792
Gemma 4 31B	Human.	250	11	0.737	0.564	0.889
Gemma 4 31B	Social	250	23	0.806	0.703	0.904
Gemma 4 31B	Formal	250	9	0.812	0.638	0.981
Gemma 4 31B	Science	250	12	0.710	0.558	0.856
Gemma 3 27B	Applied	248	67	0.585	0.526	0.646
Gemma 3 27B	Factual	250	37	0.674	0.599	0.748
Gemma 3 27B	Human.	249	24	0.573	0.488	0.665
Gemma 3 27B	Social	250	54	0.528	0.468	0.583
Gemma 3 27B	Formal	250	76	0.569	0.500	0.634
Gemma 3 27B	Science	250	63	0.598	0.543	0.645
Gemma 3 4B	Applied	244	118	0.536	0.485	0.585
Gemma 3 4B	Factual	241	65	0.564	0.494	0.637
Gemma 3 4B	Human.	240	55	0.506	0.441	0.562
Gemma 3 4B	Social	240	113	0.488	0.430	0.543
Gemma 3 4B	Formal	239	131	0.541	0.483	0.609
Gemma 3 4B	Science	241	112	0.566	0.507	0.621
Gemma 3 12B	Applied	242	77	0.533	0.463	0.601
Gemma 3 12B	Factual	237	47	0.614	0.522	0.696
Gemma 3 12B	Human.	243	29	0.522	0.420	0.625
Gemma 3 12B	Social	247	71	0.464	0.392	0.537
Gemma 3 12B	Formal	233	99	0.554	0.502	0.605
Gemma 3 12B	Science	244	66	0.500	0.430	0.575
Gemma 3 1B	Applied	249	163	0.495	0.465	0.524
Gemma 3 1B	Factual	250	124	0.501	0.472	0.531
Gemma 3 1B	Human.	250	112	0.500	0.476	0.523
Gemma 3 1B	Social	250	161	0.490	0.463	0.514
Gemma 3 1B	Formal	250	185	0.503	0.477	0.527
Gemma 3 1B	Science	250	172	0.502	0.479	0.520
GPT-oss-120B	Applied	93	18	0.829	0.742	0.902
GPT-oss-120B	Factual	65	7	0.855	0.625	0.989
GPT-oss-120B	Human.	81	5	0.611	0.244	0.925
GPT-oss-120B	Social	78	11	0.801	0.659	0.923
GPT-oss-120B	Formal	99	2	0.515	0.128	0.918
GPT-oss-120B	Science	102	11	0.760	0.582	0.900
GPT-oss-20B	Applied	250	57	0.769	0.697	0.827
GPT-oss-20B	Factual	250	34	0.766	0.675	0.847
GPT-oss-20B	Human.	250	25	0.803	0.688	0.904

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Table A6 – continued

Model	Domain	n	Errors	AUROC	CI lo	CI hi
GPT-oss-20B	Social	250	27	0.827	0.745	0.901
GPT-oss-20B	Formal	250	16	0.758	0.588	0.896
GPT-oss-20B	Science	250	20	0.709	0.592	0.810
GPT-5.4 mini	Applied	227	45	0.743	0.649	0.821
GPT-5.4 mini	Factual	216	26	0.797	0.712	0.870
GPT-5.4 mini	Human.	230	14	0.723	0.598	0.834
GPT-5.4 mini	Social	228	55	0.604	0.508	0.693
GPT-5.4 mini	Formal	232	64	0.572	0.492	0.649
GPT-5.4 mini	Science	223	41	0.725	0.631	0.815
GPT-5.4	Applied	250	29	0.818	0.735	0.887
GPT-5.4	Factual	250	21	0.777	0.652	0.880
GPT-5.4	Human.	250	11	0.640	0.397	0.876
GPT-5.4	Social	250	27	0.747	0.638	0.849
GPT-5.4	Formal	250	41	0.539	0.452	0.631
GPT-5.4	Science	250	28	0.649	0.533	0.754
GPT-5.4 nano	Applied	250	85	0.633	0.565	0.700
GPT-5.4 nano	Factual	250	58	0.749	0.674	0.813
GPT-5.4 nano	Human.	250	50	0.726	0.652	0.798
GPT-5.4 nano	Social	250	100	0.486	0.410	0.564
GPT-5.4 nano	Formal	250	132	0.548	0.477	0.620
GPT-5.4 nano	Science	250	95	0.615	0.538	0.691
Qwen Think	Applied	250	22	0.726	0.611	0.838
Qwen Think	Factual	250	18	0.680	0.528	0.811
Qwen Think	Human.	250	10	0.714	0.529	0.875
Qwen Think	Social	250	32	0.768	0.695	0.836
Qwen Think	Formal	250	8	0.644	0.443	0.838
Qwen Think	Science	250	14	0.757	0.630	0.877
Qwen Coder	Applied	250	49	0.650	0.585	0.716
Qwen Coder	Factual	250	29	0.716	0.637	0.791
Qwen Coder	Human.	250	16	0.723	0.597	0.827
Qwen Coder	Social	250	51	0.616	0.540	0.688
Qwen Coder	Formal	250	23	0.576	0.459	0.690
Qwen Coder	Science	250	28	0.696	0.603	0.794
Qwen 80B Inst	Applied	250	45	0.647	0.571	0.723
Qwen 80B Inst	Factual	249	28	0.667	0.571	0.770
Qwen 80B Inst	Human.	250	12	0.716	0.564	0.854
Qwen 80B Inst	Social	250	49	0.691	0.621	0.759
Qwen 80B Inst	Formal	250	46	0.585	0.516	0.671
Qwen 80B Inst	Science	249	35	0.584	0.483	0.680
Qwen 235B	Applied	250	55	0.644	0.581	0.711
Qwen 235B	Factual	250	23	0.622	0.522	0.728
Qwen 235B	Human.	250	16	0.711	0.601	0.827
Qwen 235B	Social	250	33	0.650	0.538	0.740
Qwen 235B	Formal	250	42	0.567	0.478	0.655
Qwen 235B	Science	250	32	0.637	0.573	0.695
GLM-5	Applied	84	4	0.602	0.221	0.916
GLM-5	Factual	111	8	0.576	0.399	0.775
GLM-5	Human.	91	3	0.818	0.706	0.972
GLM-5	Social	100	14	0.790	0.648	0.908
GLM-5	Formal	102	3	0.609	0.388	1.000
GLM-5	Science	110	3	0.554	0.387	0.853